New Uzbekistan University

Statement of Ethics: I agree to complete this exam without unauthorized assistance.

FULL NAME: _____ SIGNATURE: _____

ID NUMBER: _____ EXAM ROOM: _____ GROUP: _____

Mock Midterm Exam

COURSE NAME: Computer Programming I COURSE NUMBER: CS 102

EXAMINATION DURATION: 90 MINUTES EXAM VERSION MOCK

ADDITIONAL MATERIALS ALLOWED TO USE: No additional materials or items are allowed to be used.

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST! VIOLATION OF THE RULES CAN LEAD TO A LOSS OF POINTS.

- There are **7 problems** worth **100 points**. Write every answer **in the space provided**; if you need more space, use the back of the page and say so.
- Assume every code fragment is inside `main` with `#include <iostream>`, `<iomanip>`, `<cmath>` and `using namespace std;`. When a fragment shows its own functions, the lines after `// in main:` are inside `main`.
- Points are shown in brackets — roughly one minute per point. Partial credit is given for clear reasoning. Small syntax slips in your code cost little; wrong logic costs a lot.
- Put your signature and write your ID on every page.
- Follow the University Examination Rules.

FOR INSTRUCTOR USE ONLY (DO NOT WRITE ANYTHING):

Problem	Type of Questions	Points	Score
1	Short Answer	14	
2	Number Systems and Data	12	
3	Algorithms	12	
4	What Does It Print?	20	
5	Debugging	12	
6	Complete the Program	10	
7	Write the Code	20	
Total		**100**	

SIGNATURE: _____ ID: _____

Problem 1: Short Answer **(14 points)**

For (a)–(e) circle **TRUE** or **FALSE**. For (f)–(g) write a short answer on the line.

(a) [2 points] `cout << 'a' + 1;` prints `b`. TRUE FALSE

(b) [2 points] `for (int i = 0; i < 10; i++); { cout << i; }` compiles and prints `10`. TRUE FALSE

(c) [2 points] A function `void f(int& x)` can be called as `f(3 + 4)`. TRUE FALSE

(d) [2 points] In C++, `-7 / 2` is `-3` and `-7 % 2` is `-1`. TRUE FALSE

(e) [2 points] `int n = 3.99;` stores the value 4 in `n`. TRUE FALSE

(f) [2 points] After this code, what is the value of `count`? Show how you counted.

```
int count = 0; for (int i = 1; i <= 20; i *= 2) for (int j = i; j <= 20; j += i) count++;
```

_____

(g) [2 points] Why does `for (double x = 0; x != 1.0; x += 0.1)` never stop? How should the condition be written?

_____

Problem 2: Number Systems and Data **(12 points)**

Show your work for (a)–(d).

(a) [2 points] Compute `0x1F4 + 0b1011` in decimal and in binary. _____

(b) [2 points] Convert $\texttt{3A7}_{16}$ to octal (hint: go through binary). _____

(c) [2 points] Convert 1234_5 (base 5) to decimal and then to octal. _____

(d) [2 points] How many bits does $100\,000$ need? What is the largest number with that many bits? _____

(e) [2 points] What does this print? (`'a'` is 97 in ASCII.)

```
char c = 'z';
for (int i = 0; i < 3; i++) c = (c - 'a' + 5) % 26 + 'a';
cout << c << (int)c;
```

_____

(f) [2 points] `int a = 100000; long long b = a * a; long long c = (long long)a * a;` What are `b` and `c`, and why are they different?

_____

Problem 3: Algorithms: Pseudocode, Tracing, Flowcharts **(12 points)**

(a) [6 points] Trace this pseudocode: one row per pass (values at the *end* of the pass). Write what it prints, and say in one sentence what `r` contains for any input.

```
SET n = 73582
SET r = 0
SET c = 0
WHILE n > 0 DO
    SET d = n MOD 10
    IF d MOD 2 = 0 THEN
        SET r = r * 10 + d
    ELSE
        SET c = c + 1
    SET n = n DIV 10
PRINT r, c
```

Pass	d	n	r	c
before	—	73582	0	0
1				
2				
3				
4				
5				

Prints: _____

`r` is: _____

(b) [3 points] This pseudocode should print how many digits of n are equal to its *last* digit (for 2522 it must print 3). It contains **two** faults: find and correct both.

```
INPUT n
SET last = n MOD 10
SET count = 0
WHILE n > 9 DO
    IF n MOD 10 = last THEN
        SET count = count + 1
    SET n = n MOD 10
PRINT count
```

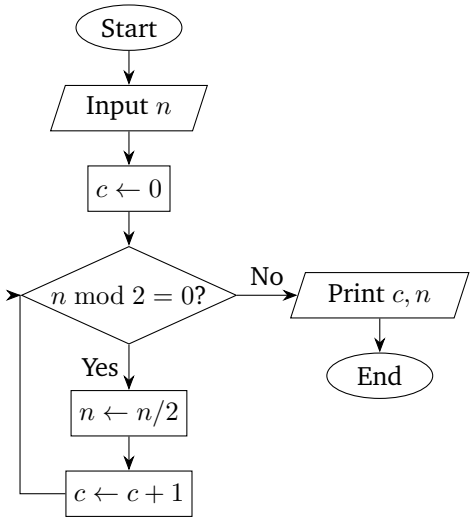
Fault 1: _____

Fix 1: _____

Fault 2: _____

Fix 2: _____

(c) [3 points] What does this flowchart print for $n = 96$? For $n = 40$? What happens for $n = 0$, and why?



$n = 96$: _____

$n = 40$: _____

$n = 0$: _____

Problem 4: What Does It Print? **(20 points)**

Write **exactly** what each fragment prints (mind the spaces). If a fragment does **not compile**, write `ERROR` and explain why.

(a) [4 points]

```
int a = 5, b = 2;
double x = a / b * 1.0 + (double)(a / b) + a / (double)b;
int y = a++ * 2 + --b;
bool z = !(a > b) || (b = 0);
cout << x << " " << y << " " << a << " " << b << " " << z << endl;
```

Output: _____

(b) [4 points]

```
char op = 'b';
int v = 10;
switch (op + 1) {
    case 'a': v -= 1;
    case 'c': v *= 3;
    case 'd': v = v > 20 ? v - 20 : v + 20;
    default:  v %= 7;
}
cout << v << endl;
```

Output: _____

(c) [4 points]

```
int s = 0;
for (int i = 1; i <= 5; i++) {
    int j = i;
    while (j > 0) {
        j -= 2;
        if (j == 1) continue;
        if (i % 4 == 0) break;
        s += i * j;
    }
}
cout << s << endl;
```

Output: _____

(d) [4 points]

```
void swapIf(int& a, int& b) {
    if (a > b) { int t = a; a = b; b = t; }
}
// in main:
int x = 9;
swapIf(x, 4);
cout << x << endl;
```

Output: _____

(e) [4 points]

```
int g(int& a, int b = 3) {
    for (int i = 0; i < b; i++) a += i;
    return a * b;
}
int g(double a) {
    return a * 10;
}
// in main:
int p = 2, q = 2;
int r1 = g(p);
int r2 = g(q, p);
cout << r1 << " " << r2 << " " << p << " " << q << " " << g(p / 4.0) << endl;
```

Output: _____

Problem 5: Debugging **(12 points)**

This program should read two numbers a and b (in **any** order) and print how many numbers between them (both included) have **strictly increasing** digits from left to right (like 7, 25, 389 — but not 22 or 31), and the **sum** of those numbers. For the input `20 10` it must print `8 124` (12, 13, . . . , 19). It contains exactly **four** bugs; the program compiles, so all four are logic errors. For each bug give the line number(s), what is wrong, and the corrected code.

(a) [12 points]

```
1  #include <iostream>
2  using namespace std;
3
4  bool increasing(int n);
5
6  int main() {
7      int a, b, count = 0, sum = 0;
8      cin >> a >> b;
9      if (a > b) {
10         a = b;
11         b = a;
12     }
13     for (int x = a; x <= b; x++) {
14         if (increasing(x)) {
15             count++;
16             sum = x;
17         }
18     }
19     cout << count << " " << sum << endl;
20     return 0;
21 }
22
23 bool increasing(int n) {
24     int prev = 0;
25     while (n > 0) {
26         int d = n % 10;
27         if (d > prev) {
28             return false;
29         }
30         prev = d;
31         n /= 10;
32     }
33     return true;
34 }
```

Line	What is wrong	Corrected code

Problem 6: Complete the Program **(10 points)**

The program reads $n \geq 0$ and a base b (2 to 9) and prints n written in base b, and how many digits that takes. The result is built as a number whose decimal digits are the base-b digits. Input `2026 8` must print `3752 (4 digits)`; input `0 2` must print `0 (1 digits)`. Fill in the five gaps.

(a) [10 points]

```
void toBase(int n, int base, (1)_____ result, int& digits);

int main() {
    int n, base;
    cin >> n >> base;
    long long r;
    int k;
    (2)_____;
    cout << r << " (" << k << " digits)" << endl;
    return 0;
}

void toBase(int n, int base, long long& result, int& digits) {
    result = 0;
    digits = 0;
    long long place = 1;
    do {
        result += (3)_____;
        place (4)_____;
        n (5)_____;
        digits++;
    } while (n > 0);
}
```

(1) [ ] (2) [ ] (3) [ ] (4) [ ] (5) [ ]

Problem 7: Write the Code **(20 points)**

(a) [10 points] Write three functions. `int countDigits(int n)` returns the number of digits of $n \geq 0$. `int rotate(int n)` moves the *last* digit of n to the front (`rotate(1234)` is 4123, `rotate(98)` is 89, `rotate(7)` is 7). `bool isRotation(int a, int b)` returns `true` when b can be obtained from a by rotating it some number of times (1234 and 3412: true; 1234 and 1243: false). `rotate` and `isRotation` must **use** the functions above them. You may assume the numbers contain no digit 0. Write **only** the three function definitions; for full marks, do not use `string`, `to_string`, `pow` or any other built-in/library function.

(b) [10 points] Write a complete program (`main` only). It reads n (you may assume $1 \leq n \leq 9$) and prints the numbers $1, 2, \ldots, n^2$ in an $n \times n$ *snake*: the first row goes left to right, the second right to left, the third left to right again, and so on. Every number takes 4 columns (`setw(4)`). Finally print the sum of the main diagonal (top left to bottom right). Use nested `for` loops and compute every number from its row and column — no arrays.

Input 4:

```
   1   2   3   4
   8   7   6   5
   9  10  11  12
  16  15  14  13
Diagonal: 32
```

END OF EXAM